

Radiative Transitions of Strange Heavy Axial Mesons from Light-Cone QCD Sum Rules

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Abstract

We present a light-cone QCD sum-rule analysis of $D_{s1}(2460) \rightarrow D_s \gamma$, $D_{s1}(2536) \rightarrow D_s \gamma$, and $B_{s1}(5830) \rightarrow B_s \gamma$. The calculation keeps explicit strange-quark-mass terms and includes hard photon emission from both quark lines, two-particle photon distribution amplitudes, and the three-particle photon-amplitude corrections induced by the background-gluon term in the quark propagator. We separate the basis-current sum rules from the physical-current residue normalization. For the charm-strange channel the mixed-current residue model is calibrated by a two-point input and checked against a pure axial-current benchmark; for the bottom-strange channel we quote a Pullin-Zwicky decay-constant scenario. Using the modern lattice normalization of the leading-twist photon distribution amplitude, we find $\Gamma[D_{s1}(2460) \rightarrow D_s \gamma] = 16.1^{+4.4}_{-3.7}$ keV, $\Gamma[D_{s1}(2536) \rightarrow D_s \gamma] = 0.504^{+0.600}_{-0.377}$ keV, and a sub-keV median prediction for the observed $B_{s1}(5830) \rightarrow B_s \gamma$ channel.

1 Introduction

Radiative transitions of positive-parity strange heavy mesons probe the interplay of heavy-quark spin symmetry, light-quark dynamics, and threshold effects. The two charm-strange axial states, $D_{s1}(2460)$ and $D_{s1}(2536)$, are especially useful because the lower state has a measured radiative branching fraction [1], while the upper state has a measured total width but no prominent $D_s \gamma$ signal [2]. The bottom-strange state $B_{s1}(5830)$ is observed in hadronic spectroscopy [3]; its radiative decay has not yet been measured.

The aim of this paper is not only to quote updated numbers, but also to make the normalization of the calculation unambiguous. The light-cone OPE gives transition form factors for basis currents. The conversion to the physical 1^+ states requires the residues of the physical mixed currents. We therefore organize the analysis in stages: first compute the axial and tensor basis amplitudes, then add the photon-DA corrections, and finally apply the physical-current normalization.

The hierarchy between $D_{s1}(2460) \rightarrow D_s \gamma$ and $D_{s1}(2536) \rightarrow D_s \gamma$ was recently emphasized by Bondar and by Bondar and Milstein [4, 5]. Their quark-model explanation is that the physical $J = 1$ states are mixed spin configurations and that the E1 amplitudes enter with different relative signs. Below we show that the same mechanism appears directly in the LCSR decomposition of the physical amplitudes.

Two photon-input scenarios are considered throughout. The first is the legacy magnetic-susceptibility normalization of the photon distribution amplitudes [6, 7]. The second uses the recent lattice normalization of the leading-twist photon distribution amplitude [8]. The latter is our preferred modern scenario; the former is kept to show continuity with older light-cone sum-rule analyses.

2 Hadronic Conventions

We use $P = p + q$, where P is the momentum of the initial axial meson, p is the momentum of the final pseudoscalar meson, and q is the real-photon momentum. The pseudoscalar interpolating current is

$$J_5 = \bar{s} i \gamma_5 Q, \quad Q = c, b. \quad (1)$$

For the axial channel we introduce two basis currents,

$$J_\mu^A = \bar{s} \gamma_\mu \gamma_5 Q, \quad (2)$$

$$J_\mu^B = \frac{i}{m_Q + m_s} \bar{s} \sigma_{\mu\alpha} P^\alpha \gamma_5 Q. \quad (3)$$

The momentum in the tensor current is the initial-state momentum P^α . The physical-current combinations are

$$J_{1\mu} = \sin \theta J_\mu^A + \cos \theta J_\mu^B, \quad (4)$$

$$J_{2\mu} = \cos \theta J_\mu^A - \sin \theta J_\mu^B, \quad (5)$$

with central value $\theta = 35.3^\circ$ and a scan over $25^\circ \leq \theta \leq 45^\circ$.

The radiative matrix element is parametrized as

$$\langle P(p) \gamma(q, \varepsilon) | A(P, \eta) \rangle = eG [(\eta \cdot \varepsilon^*)(p \cdot q) - (\eta \cdot q)(p \cdot \varepsilon^*)]. \quad (6)$$

The corresponding width is

$$\Gamma(A \rightarrow P \gamma) = \frac{\alpha_{\text{em}}}{3} G^2 \left(\frac{m_A^2 - m_P^2}{2m_A} \right)^3. \quad (7)$$

3 Light-Cone Sum Rule

The starting correlation function is

$$\Pi_\mu(p, q) = i \int d^4x e^{ip \cdot x} \langle \gamma(q, \varepsilon) | T \{ \bar{s}(x) \Gamma_\mu Q(x) \bar{Q}(0) i \gamma_5 s(0) \} | 0 \rangle, \quad (8)$$

where Γ_μ is either J_μ^A or J_μ^B . We project onto the gauge-invariant E1 tensor

$$T_{\mu\nu} = (p \cdot q) g_{\mu\nu} - q_\mu p_\nu. \quad (9)$$

The OPE is constructed with quark propagators in an external electromagnetic field. Both hard photon emission and soft photon distribution-amplitude terms are retained. In addition, the background-gluon insertion in the propagator generates the three-particle photon-DA terms. The ordinary vacuum-condensate pieces of the propagator are not inserted as separate three-point corrections: in the photon LCSR they enter through photon matrix elements and normalizations such as $\chi_s \langle \bar{s}s \rangle$. This is separate from the local two-point sum rules used to determine decay constants, where ordinary SVZ condensates are part of the OPE.

After Borel transformation and continuum subtraction the basis-current amplitudes are denoted G_A and G_B . The charm physical couplings are then formed as

$$G_{2460} = \frac{\sin \theta f_A G_A + \cos \theta f_B G_B}{f_1}, \quad (10)$$

$$G_{2536} = \frac{\cos \theta f_A G_A - \sin \theta f_B G_B}{f_2}. \quad (11)$$

Here f_1 and f_2 are the residues of the physical mixed currents J_1 and J_2 . We calibrate the off-diagonal two-point overlap from the working input $f_1 = 0.345 \pm 0.017$ GeV and use the same overlap to predict $f_2 \simeq 0.38$ GeV.

This residue prescription is checked against the pure axial-current benchmark of Wang, $f_A(D_{s1}) = 0.245 \pm 0.017$ GeV [9]. Since Wang’s current is the pure axial current J_A , the comparison is meaningful only in the limiting cases in which our mixed current reduces to J_A : $J_1 = J_A$ at $\theta = 90^\circ$, or $J_2 = J_A$ at $\theta = 0^\circ$. Our working pure-axial input, $f_A = 0.225 \pm 0.025$ GeV, is consistent with this benchmark within the combined uncertainty. The difference in central values can naturally come from quark-mass choices, condensate inputs, Borel windows, continuum thresholds, OPE truncation, and normalization conventions. It is therefore a useful basis-scale check, not a direct determination of the physical mixed-current residues at $\theta = 35.3^\circ$.

For the bottom-strange channel we use the Pullin–Zwicky values $f_{B_{s1}}^{\text{PZ}} = 0.305 \pm 0.027$ GeV and $f_{B_{s1}}^{T,\text{PZ}} = 0.285 \pm 0.027$ GeV [10]. The bottom quantity reported as f_{eff} below is an effective normalization entering the chosen PZ scenario. It is not a separately extracted local two-point residue f_1 or f_2 .

4 Numerical Inputs and Scale Convention

Table 1 lists the main inputs. Hadron masses, quark masses, charges, and α_{em} are taken from PDG summaries [11]. The pseudoscalar decay constants are taken in the lattice context summarized by FLAG [12]. Photon DAs follow Ball, Braun and Kivel [6]; the legacy susceptibility comparison follows Rohrwild [7]; the modern photon normalization follows the lattice result of Bacchio et al. [8]. The input bookkeeping is kept in `inputs_table.csv` and in the generated citation map `outputs/input_citation_map.csv`.

The photon-DA normalizations are quoted at $\mu = 1$ GeV, while the quark masses are the usual $\overline{\text{MS}}$ inputs, e.g. $m_s(2 \text{ GeV})$ and $m_c(m_c)$. This is a standard phenomenological convention rather than a fully matched single-scale setup. We therefore treat the lattice photon-normalized scenario as the preferred modern input and retain the susceptibility scenario as a comparison with older analyses.

Quantity	Symbol	Value	Source/use
quark masses	m_c, m_b, m_s	1.27(2), 4.18(3), 0.093(11) GeV	PDG; explicit m_s terms kept
charm masses	$M_{2460}, M_{2536}, m_{D_s}$	2.4595, 2.53511, 1.96835 GeV	PDG
bottom masses	M_{5830}, m_{B_s}	5.82870, 5.36692 GeV	PDG/CMS
lower B_{s1} diagnostic mass	$M_{B_{s1}}^{\text{low}}$	5.750(26) GeV	lattice-inspired diagnostic [13]
pseudoscalar decay constants	f_{D_s}, f_{B_s}	0.2499, 0.2303 GeV	FLAG/lattice context
charm mixed residues	f_1, f_2	0.345(17), $\simeq 0.38$ GeV	Stage-3 calibration
pure axial benchmark	$f_A(D_{s1})$	0.245(17) GeV	Wang pure-current check
bottom PZ constants	$f_{B_{s1}}^{\text{PZ}}, f_{B_{s1}}^{T,\text{PZ}}$	0.305(27), 0.285(27) GeV	bottom scenario
legacy photon input	$\chi_s(1 \text{ GeV})$	$+3.15(30) \text{ GeV}^{-2}$	comparison scenario
lattice photon input	$f_{\gamma,s}^\perp(1 \text{ GeV})$	$-55.1(1.9) \text{ MeV}$	preferred scenario

Table 1: Main numerical inputs used in the analysis.

5 Results

The final predictions are summarized in Table 3. We quote medians and 16–84 percentile intervals rather than Gaussian means and standard deviations. This is important because the $D_{s1}(2536)$ and $B_{s1}(5830)$ amplitudes can pass close to zero; their width distributions are therefore non-Gaussian.

The preferred lattice-normalized result for $D_{s1}(2460)$ is a tens-of-keV partial width. In contrast, $D_{s1}(2536)$ is highly cancellation sensitive: the same axial and tensor basis amplitudes

Case	Current/target	θ	ρ_{AB}	f_A	f_1	f_2
Wang benchmark	pure J_A current	–	–	0.245(17)	–	–
pure- J_1 axial limit	$J_1 = J_A$	90°	0	0.225	0.225	0.462
pure- J_2 axial limit	$J_2 = J_A$	0°	0	0.225	0.462	0.225
physical angle, Wang-like target	target $f_1 = 0.245$	35.3°	–1.00	0.225	0.247	0.451
physical angle, Wang f_A	target $f_1 = f_A = 0.245$	35.3°	–0.957	0.245	0.245	0.462
Stage-3 anchor	target $f_1 = 0.345$	35.3°	–0.408	0.225	0.345	0.381

Table 2: Pure-current benchmark for the charm residue model. Wang’s value tests the pure J_A scale. The physical mixed-current residues require an assumption about the AA/BB overlap. Decay constants are in GeV.

Sector	Decay	Photon input	Mixing	Normalization	G (GeV $^{-1}$)	Γ (keV)
D_s	$D_{s1}(2460) \rightarrow D_s \gamma$	legacy χ_s	35.3°	$f_1 = 0.344[0.329, 0.363]$	$-0.424[-0.484, -0.360]$	$37.7^{+11.5}_{-10.5}$
D_s	$D_{s1}(2536) \rightarrow D_s \gamma$	legacy χ_s	35.3°	$f_2 = 0.380[0.339, 0.423]$	$+0.0118[+0.0014, +0.0233]$	$0.045^{+0.124}_{-0.040}$
D_s	$D_{s1}(2460) \rightarrow D_s \gamma$	lattice $f_{\gamma,s}^\perp$	35.3°	$f_1 = 0.345[0.330, 0.364]$	$-0.277[-0.312, -0.243]$	$16.1^{+4.4}_{-3.7}$
D_s	$D_{s1}(2536) \rightarrow D_s \gamma$	lattice $f_{\gamma,s}^\perp$	35.3°	$f_2 = 0.379[0.338, 0.423]$	$+0.0403[+0.0194, +0.0596]$	$0.504^{+0.600}_{-0.377}$
B_s	$B_{s1}(5830) \rightarrow B_s \gamma$	legacy χ_s	35.3°	PZ constants, $f_{\text{eff}} = 0.391[0.354, 0.428]$	$-0.0140[-0.0387, +0.0155]$	$0.109^{+0.304}_{-0.101}$
B_s	$B_{s1}(5830) \rightarrow B_s \gamma$	lattice $f_{\gamma,s}^\perp$	35.3°	PZ constants, $f_{\text{eff}} = 0.393[0.354, 0.427]$	$+0.0356[+0.0029, +0.0694]$	$0.272^{+0.752}_{-0.244}$
B_s	$B_{s1}(5830) \rightarrow B_s \gamma$	legacy χ_s	$25^\circ\text{--}45^\circ$	PZ constants, $f_{\text{eff}} = 0.394[0.356, 0.430]$	$-0.0126[-0.0527, +0.0364]$	$0.249^{+0.570}_{-0.226}$
B_s	$B_{s1}(5830) \rightarrow B_s \gamma$	lattice $f_{\gamma,s}^\perp$	$25^\circ\text{--}45^\circ$	PZ constants, $f_{\text{eff}} = 0.387[0.348, 0.423]$	$+0.0330[-0.0228, +0.0974]$	$0.429^{+1.665}_{-0.401}$
B_s	$B_{s1}^{\text{low}}(5750) \rightarrow B_s \gamma$	legacy χ_s	35.3°	PZ constants, $f_{\text{eff}} = 0.384[0.347, 0.421]$	$+0.337[+0.279, +0.411]$	$14.2^{+7.0}_{-5.2}$
B_s	$B_{s1}^{\text{low}}(5750) \rightarrow B_s \gamma$	lattice $f_{\gamma,s}^\perp$	35.3°	PZ constants, $f_{\text{eff}} = 0.386[0.350, 0.424]$	$+0.474[+0.420, +0.546]$	$27.8^{+11.6}_{-7.7}$

Table 3: Final form-factor and width predictions. The lattice photon input is the preferred modern normalization; the legacy susceptibility input is kept as a comparison with older photon-DA analyses. The $B_{s1}^{\text{low}}(5750)$ entries are lower-partner diagnostics, not observed resonances.

enter with the opposite physical combination in Eq. (11), driving the amplitude close to zero for a wide part of the accepted input range. The observed $B_{s1}(5830)$ behaves similarly in the bottom sector. Its median width is sub-keV in the PZ decay-constant scenario, although the mixing-angle scan leaves a long upper tail.

Table 4 makes this cancellation explicit. The entries are the two normalized pieces of Eq. (11). For $D_{s1}(2460)$ both pieces have the same sign and add constructively. For $D_{s1}(2536)$ they have opposite signs and nearly cancel. This is the LCSR version of the Bondar–Milstein explanation of why the $D_{s1}(2536)$ radiative mode is difficult to observe.

Photon input	Mixing	A_{2460}	B_{2460}	G_{2460}	A_{2536}	B_{2536}	G_{2536}	$ G_{2460}/G_{2536} $
legacy χ_s	35.3°	–0.135	–0.287	–0.424	–0.174	+0.184	+0.0118	36.1
lattice $f_{\gamma,s}^\perp$	35.3°	–0.0715	–0.205	–0.277	–0.0917	+0.132	+0.0403	6.9

Table 4: Median normalized axial and tensor pieces of the physical amplitudes. The 2460 pieces are constructive, whereas the 2536 pieces are destructive. All G values are in GeV $^{-1}$.

Table 5 places the preferred photon-normalized results next to representative quark-model and phenomenological predictions collected by Bondar and Milstein. For channels with a vector final state, $D_s^* \gamma$ or $B_s^* \gamma$, we quote the literature values but leave our entry blank because those transitions require separate LCSR correlators and are not part of the present calculation.

For complementarity with the bottom-sector discussion of Ref. [5], we also keep a lower-partner diagnostic in the Stage-3 scan. This row uses a model mass near 5.75 GeV and should not be described as an experimentally established state. It is nevertheless useful because Bondar and Milstein quote both a lower $B_{s1}(5748)$ and the observed high state near 5.829 GeV. In our PZ-normalized LCSR scan the lower diagnostic state has

$$\Gamma[B_{s1}^{\text{low}} \rightarrow B_s \gamma] = 14.2[8.9, 21.2] \text{ keV}$$

Channel	Early QM	Rel. QM	Recent QM	Bondar–Milstein	This work
$D_{s1}(2536) \rightarrow D_s \gamma$	15 [14]	25.2–31.1 [15]	18.18–18.85 [16]	12 [5]	0.504[0.127, 1.104]
$D_{s1}(2460) \rightarrow D_s \gamma$	6.2 [14]	10.3–17.2 [15]	3.53–3.61 [16]	297 [5]	16.1[12.4, 20.5]
$B_{s1}^{\text{low}}(5750) \rightarrow B_s \gamma$	37 [17]	70.6 [18]	97.7 [19]	47 [5]	27.8[20.1, 39.4]
$B_{s1}(5830) \rightarrow B_s \gamma$	27 [17]	47.8 [18]	56.6 [19]	28 [5]	0.272[0.028, 1.024]

Table 5: Representative literature comparison for the channels computed in this work. Widths are in keV. Our entries use the preferred lattice photon normalization. A complete machine-readable table, including the uncomputed vector-final-state channels quoted in Ref. [5], is stored in `outputs/bondar_literature_comparison_table.csv`.

for the legacy susceptibility input and

$$\Gamma[B_{s1}^{\text{low}} \rightarrow B_s \gamma] = 27.8[20.1, 39.4] \text{ keV}$$

for the lattice photon input. These values are much larger than our observed $B_{s1}(5830)$ median widths and are closer in scale to the potential-model expectations for a lower bottom-strange axial state. The difference reinforces that the observed high-combination state is the cancellation-sensitive one in our setup.

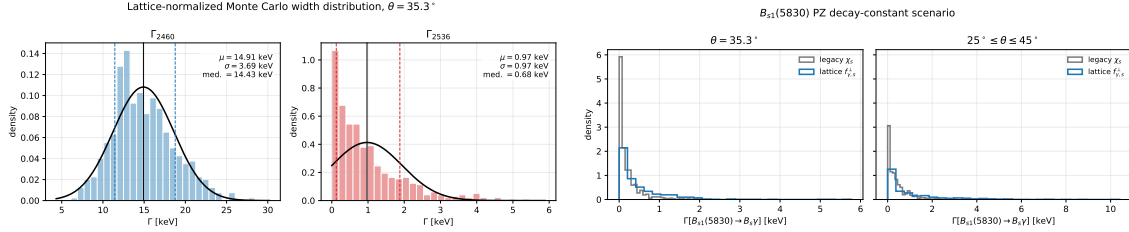


Figure 1: Monte Carlo width distributions. Left: charm-strange lattice-normalized photon input. Right: bottom-strange PZ decay-constant scenario.

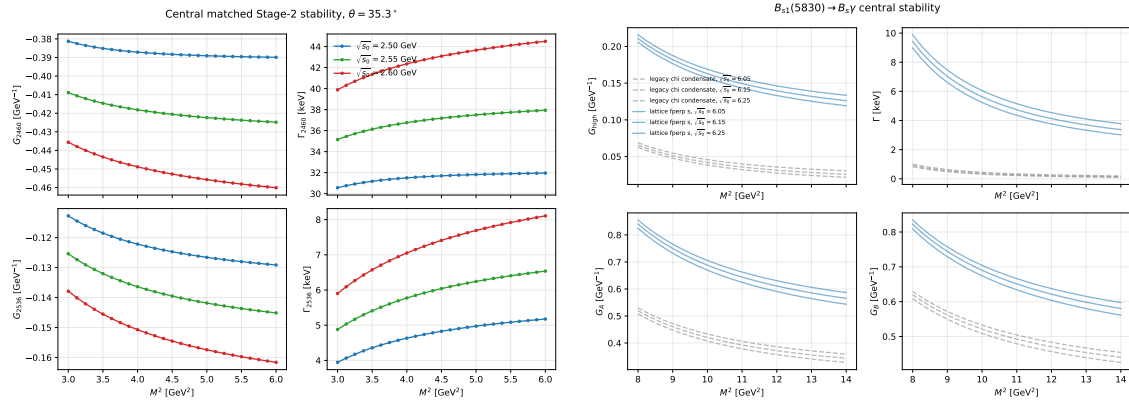


Figure 2: Representative Borel and continuum-threshold stability checks for the charm-strange and bottom-strange channels.

6 Comparison with Experiment

The theory result is a partial radiative width. Experiments may quote a branching fraction, a total width, or only spectroscopy in hadronic channels. Table 6 gives the corresponding interpretation for each available case.

Decay	Theory input	Experimental information	Interpretation
$D_{s1}(2460) \rightarrow D_s \gamma$	legacy χ_s : 37.7[27.2, 49.2] keV	$\mathcal{B} = (16 \pm 4 \pm 3)\%$ [1]	implies $\Gamma_{\text{tot}} \sim 0.24$ MeV
$D_{s1}(2460) \rightarrow D_s \gamma$	lattice $f_{\gamma,s}^\perp$: 16.1[12.4, 20.5] keV	$\mathcal{B} = (16 \pm 4 \pm 3)\%$ [1]	implies $\Gamma_{\text{tot}} \sim 0.10$ MeV
$D_{s1}(2536) \rightarrow D_s \gamma$	legacy χ_s : 0.045[0.0046, 0.169] keV	$\Gamma_{\text{tot}} = 0.92 \pm 0.05$ MeV [2]	$\mathcal{B}_\gamma \sim 0.005\%$
$D_{s1}(2536) \rightarrow D_s \gamma$	lattice $f_{\gamma,s}^\perp$: 0.504[0.127, 1.104] keV	$\Gamma_{\text{tot}} = 0.92 \pm 0.05$ MeV [2]	$\mathcal{B}_\gamma \sim 0.05\%$
$B_{s1}(5830) \rightarrow B_s \gamma$	legacy χ_s : 0.109[0.008, 0.413] keV	no direct radiative measurement [3]	future-search prediction
$B_{s1}(5830) \rightarrow B_s \gamma$	lattice $f_{\gamma,s}^\perp$: 0.272[0.028, 1.024] keV	no direct radiative measurement [3]	future-search prediction

Table 6: Interpretation of the predicted partial radiative widths against available experimental information.

The $D_{s1}(2460)$ prediction is compatible with a narrow total width when the measured branching fraction is used literally. For $D_{s1}(2536)$, the inferred radiative branching fraction is far below the percent level, which explains why this mode is difficult to observe. The $B_{s1}(5830)$ rows are predictions for future radiative searches rather than comparisons with existing radiative data.

Other effects can further reduce the visibility of $D_{s1}(2536) \rightarrow D_s \gamma$: the radiative branching fraction is diluted by the measured total width, the photon has to be isolated in an inclusive environment, and the amplitude sits near a cancellation point, making the signal sensitive to mixing and normalization details. These experimental and phenomenological issues are consistent with the upper-limit discussion of Ref. [4].

7 Reproducibility

Table 7 records the source of each headline table and figure. The machine-readable version is `outputs/manuscript_reproducibility_map.csv`.

Manuscript item	Source script	Source output	Purpose
input table	manual bookkeeping	<code>inputs_table.csv</code>	input values and citations
residue benchmark	<code>stage3_residue_benchmark_checks.py</code>	<code>stage3_residue_benchmark_checks.csv</code>	pure-current validation
main result table	Stage-3 charm and bottom scripts	<code>combined_recommended_results_table.csv</code>	final G and Γ values
experimental table	manual summary	<code>experimental_comparison_table.csv</code>	comparison with measured information
MC width plots	lattice/PZ scan scripts	<code>lattice_fperp_...pdf</code> , <code>stage3_bs1_...pdf</code>	uncertainty distributions
stability plots	stability scan scripts	<code>stage2_stability_central.pdf</code> , <code>bs1_5830_stability.pdf</code>	Borel and threshold checks
validation checks	validation and Ward scripts	<code>validation_checks_summary.txt</code> , <code>ward_transversality_check.txt</code>	limiting cases and gauge structure
interference diagnostic	<code>ds1_interference_diagnostic.py</code>	<code>ds1_interference_diagnostic_summary.csv</code>	constructive/destructive amplitude pieces

Table 7: Reproducibility map for the headline material in this draft.

8 Conclusions

We have computed the radiative transitions of strange heavy axial mesons in a staged light-cone QCD sum-rule framework. The calculation includes hard photon emission from both quark lines, two-particle and three-particle photon distribution amplitudes, and explicit strange-quark-mass terms. The charm residue normalization has been separated from the basis-current OPE and checked against the pure axial-current benchmark. The bottom result is quoted as a PZ-normalized decay-constant scenario.

The physical pattern is robust. The $D_{s1}(2460)$ radiative mode can naturally lie at the tens-of-keV level. The $D_{s1}(2536)$ mode is strongly suppressed by the axial–tensor cancellation, and the observed $B_{s1}(5830)$ radiative width is predicted to have a sub-keV median. These suppressions are consistent with the current experimental situation and provide concrete targets for future radiative searches.

Acknowledgments

To be added.

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